hw12

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Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
data <- read.table("https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000

head(data)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```
summary(data)
```

```
    sample              geno                exp
 Length:462         Length:462         Min.   : 6.675
 Class :character   Class :character   1st Qu.:20.004
 Mode  :character   Mode  :character   Median :25.116
                                       Mean   :25.640
                                       3rd Qu.:30.779
                                       Max.   :51.518
```

```
table(data$geno)
```

```

A/A A/G G/G
108 233 121
```

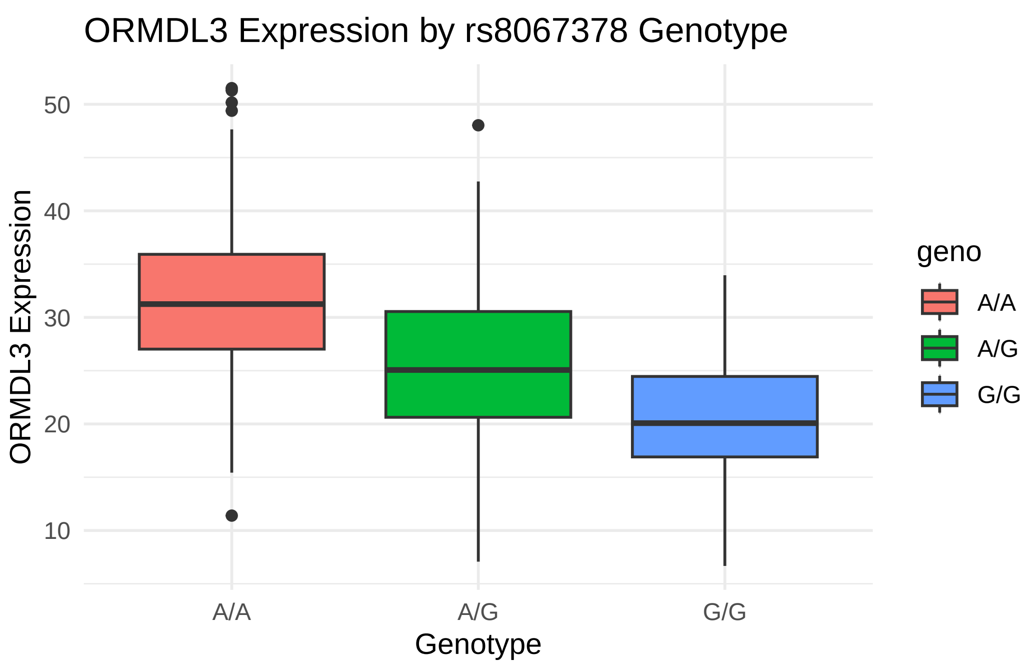
```r
tapply(data$exp, data$geno, median)
```

```
     A/A      A/G      G/G 
31.24847 25.06486 20.07363 
```

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```r
library(ggplot2)

ggplot(data, aes(x = geno, y = exp, fill = geno)) +
  geom_boxplot() +
  labs(
    title = "ORMDL3 Expression by rs8067378 Genotype",
    x = "Genotype",
    y = "ORMDL3 Expression"
  ) +
  theme_minimal()
```



ORMDL3 expression order is A/A, A/G, and G/G, from highest to lowest. This suggests that rs8067378 is associated with ORMDL3 expression, presence of G allele being correlated to lower expression levels. Therefore, the SNP affects the expression of ORMDL3.